Teaching Statement

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I have always been passionate about teaching, which I view as an integral part of an academic career. Teaching and mentoring, to me, is an opportunity to inspire curiosity, creativity, and confidence in students. My approach is grounded in the same philosophy that drives my research – learning by doing. I have viewed courses as not merely a medium of transferring knowledge but also as a platform to develop the students' ability to think critically and be able to apply the knowledge in realistic settings via a hands-on approach. Similarly, when mentoring students, ranging from PhD to undergrads, because of my deeply experiential research nature, I encourage students to approach problems with curiosity and a hands-on mindset, allowing them to move beyond theory and engage with the systems they study. Through this process, I aim to instill a passion for exploration and a commitment to excellence in their future pursuits.

Teaching Experience

During my Ph.D. journey, I have primarily served as a teaching assistant for two courses — *Fundamentals of Computer Networks/Modern Networking Concepts (EECE 7374, CSE 489-589) and Mobile and Wireless Networks (EECE 7364)*. Among these, *CSE 489-589/EECE 7374* is the course I have TA'd most frequently, almost every academic year. This foundational networking course introduces students to the core principles of computer networks. The course includes homework assignments, Wireshark-based labs, and programming projects on socket programming and reliable data transport. I was responsible for deploying Python- and C/C++-based programming assignments, grading labs and homework, and assisting students in debugging their assignments while reinforcing their conceptual understanding. Beyond teaching technical content, this course exposed me to a diverse mix of students from both CS and ECE backgrounds, which highlighted the importance of flexibility in course design. For instance, while CS students were generally comfortable with programming, many ECE students required an introductory session before starting the assignments. To bridge this gap, I helped design and deliver short programming primers, ensuring that all students could engage meaningfully with the course material regardless of their prior experience.

EECE 7364 (Mobile and Wireless Networks) is a graduate-level course that covers the state of the art in wireless and mobile networking from a practical, systems-oriented perspective. The course material combines both current practices and cutting-edge research, spanning single-hop networks (such as Wi-Fi–based WLANs and cellular systems) and multi-hop networks (including ad hoc, mesh, and LEO satellite systems). This was a relatively new course introduced by my PhD advisor and became one of my most rewarding teaching experiences. As a TA, I played a central role in developing the lab component from the ground up. We integrated an Open Air Interface (OAI) box as a base station for the first time at Northeastern University, enabling students to directly interact with different layers of a cellular network. I along with another TA designed the lab structure, created exercises and quizzes, and guided students as they tackled real-world networking challenges. Watching their enthusiasm as they connected theory with hands-on experimentation reaffirmed my belief that the most meaningful learning occurs through active engagement.

Mentoring Experience

Given the collaborative nature of experimental wireless research, mentoring has been an integral part of my academic journey. Over the years, I have mentored several undergraduate, master's, and Ph.D. students in research, including three Ph.D. students (Sizhe Wang, Yufei Feng, and Zhekun Yu), seven M.S. students, and one undergraduate student. I have guided them through all stages of research – from idea conception to data collection, analysis, and paper publication.

My mentoring philosophy is dual-faceted. First, I focus on understanding each student's research interests and existing skill set, and how these align with the goals of a project. I then encourage them to think broadly, refine their ideas, and develop the ability to reason independently about research problems. Second, because my work spans all layers of the network stack – from the physical to the application layer – and involves both systems building and large-scale experimentation, I recognize that students' strengths often vary. Some excel in experimental evaluation but lack experience in systems development, while others have the opposite profile. I strive to help students expand their expertise beyond their comfort zones, cultivating a balance of depth and breadth that I believe is essential for success in the field.

In a more formal setting, I have also served as a mentor for EECE 7398 (Special Topics: Wireless Networks System & Applications), where I advised multiple student groups as they developed research ideas and turned them into publishable papers at top venues such as ACM CoNEXT, PAM, ACM mmNets, IEEE MASS, and ACM EMS. These experiences have strengthened my belief that mentorship is not just about providing technical guidance but also about fostering curiosity, independence, and confidence in students. As a faculty member, I plan to follow the same approach in mentoring Ph.D. students, building upon my experiences as a Ph.D. student.

Courses I Aspire to Teach

Given my research background and teaching experience, I am confident in my ability to teach a broad range of courses in wireless networking and computer networks at both the undergraduate and graduate levels. At the undergraduate level, I am also well-prepared to teach core computer science courses such as Operating Systems, Programming, and Algorithms. At the graduate level, I am particularly interested in offering seminar-style courses on advanced topics in wireless and mobile networking, including IoT networks, non-terrestrial networks, cellular systems, smartphone networking, networked XR systems, and cross-layer design. In these courses, students would engage in research paper discussions from top conferences and journals and carry out independent research projects on topics of their choice.

I also plan to design hands-on lab components where students can gain practical experience working with wireless software-defined radios such as USRPs. These labs will allow them to design and evaluate systems aimed at optimizing network performance under real-world conditions. My prior experience as a teaching assistant for EECE 7364 (Mobile and Wireless Networks) has provided me with the expertise to effectively develop and instruct such courses that blend theory with hands-on experimentation.

Overall, my goal as a teacher and mentor is to equip students with the analytical and practical skills necessary to explore complex problems independently and to inspire them to approach challenges with creativity and confidence. Through this approach, I aim to nurture the next generation of engineers and researchers who will contribute meaningfully to the advancement of networking and wireless systems.